

Response to RFI: Accelerating the Adoption and Use of Artificial Intelligence as Part of Clinical Care

Public Comment Submission

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This comment is intended for public posting and does not include patient-level protected health information (PHI) or confidential business information.

It includes limited organizational contact information solely to support follow-up on technical standards and policy implementation questions.

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1. Executive Summary: Risk-tiered governance and verification can accelerate responsible AI adoption in clinical care

Governing thought: HHS can accelerate safe and equitable clinical AI adoption by making governance and auditability requirements legible, portable, and interoperable across products and sites.

Many adoption failures stem less from algorithmic novelty than from gaps in traceability (what ran, when, for whom, and under which configuration), change control, and accountable human use in workflow.

This comment recommends risk-tiered minimum governance and auditability expectations for non-device clinical AI / decision support, tailored by runtime behavior (deterministic guideline-logic artifacts vs probabilistic inference services), and implemented through HHS levers that do not alter FDA jurisdiction over medical devices.

Key message to HHS

- Publish a risk-tiered “minimum governance + auditability expectations” framework for non-device clinical AI / decision support that is tailored by runtime behavior and intended use, and explicitly does not modify FDA device jurisdiction.
- Extend (not replace) ONC’s Decision Support Interventions (DSI) transparency and intervention risk management concepts by emphasizing runtime auditability: version/configuration capture, provenance, user actions, and change logs that enable retrospective reconstruction.
- Make the “AI audit artifact package” feasible as a minimum: a small set of artifacts that can be produced, exported, and reviewed across developers, certified health IT, and deployers without requiring proprietary disclosure of model internals.
- Use ONC Certification implementation guidance and test procedures to ensure certified health IT can capture/export AI-relevant provenance and user actions for AI-influenced documentation and orders using FHIR building blocks (SMART, CDS Hooks, SDC, Provenance).
- Use CMS levers (CMMI model terms, MIPS Improvement Activities / Promoting Interoperability pathways, and hospital quality reporting guidance) to create incentives for deployers to maintain governance artifacts and post-deployment monitoring appropriate to risk tier and runtime behavior, with candidate CFR anchors provided for counsel validation.
- Certain internal logical properties of deterministic guideline-logic artifacts (e.g., consistency, reachability, completeness over defined states) can be formally verified; this does not prove clinical benefit, data quality, or appropriateness of guideline interpretation across contexts.
- Deterministic artifacts do not require statistical “model drift” monitoring, but they still require post-deployment safety monitoring for data quality/missingness, user overrides/escalations, disparate application, and guideline/version updates with regression test outcomes.
- OCR should operationalize Section 1557 requirements for patient care decision support tools (45 CFR § 92.210) through implementation guidance that links nondiscrimination duties to concrete documentation, monitoring, and mitigation steps.

About the submitter / transparency

PCare+ is a health IT and standards-focused organization that participates in interoperability implementation and governance discussions relevant to clinical decision support and AI-enabled workflow integration.

PCare+ may have professional and/or commercial interests in approaches that favor interoperable auditability and standards-based integration, and this comment is offered to support implementable HHS policy and program design rather than to advocate for any single product or vendor.

Recommendations are framed as minimum expectations that can be met by multiple technical architectures and multiple market participants.

Summary recommendation

(1) Distinguish between AI-compiled deterministic artifacts and probabilistic AI services in regulatory, evaluation, and monitoring frameworks. These are fundamentally different technologies with different risk profiles. Regulating compiled guideline logic under monitoring rules designed for chatbots imposes compliance costs without safety benefit.

(2) Adopt formal verification of logical completeness (exhaustiveness, determinism, reachability) as the evaluation standard for deterministic clinical decision logic, beyond the necessary but insufficient baseline of audit logging.

(3) Align reimbursement incentives to verified guideline-concordant care, measured by screening completion, provenance traceability, and patient outcomes, rather than to AI tool access or adoption volume.

These recommendations are complementary to the existing computable guideline, eCQM, and clinical decision support ecosystem, and are intended to improve provenance, verification of internal logic properties where appropriate, workflow integration, and accountability rather than to imply that computable CDS does not already exist.

Evidence base referenced in this submission: Our analysis of 227 million T-MSIS Medicaid claims rows, 2.99 million CDC BRFSS survey responses, and HRSA UDS data from 18,751 FQHC sites indicates that preventive screening is declining nationally, even where the infrastructure and coverage mandates to deliver it exist. Patients and caregivers need understandable disclosure and practical recourse when AI influences clinical recommendations; a decision artifact whose correctness is provable provides a stronger foundation than transparency-by-logging alone.

2. RFI Compliance Crosswalk: Responses are provided in the required question order

Exhibit 1 maps each RFI question to the corresponding response section(s) and supporting exhibits/appendices.

Exhibit 1. RFI Compliance Crosswalk

ID	RFI question	Response location	Page
Q1	1. The RFI states that there are numerous challenges that prevent the use of AI in clinical care, and that HHS is seeking input on how to address these challenges. What are the barriers and opportunities to accelerate AI adoption as part of clinical care?	Question 1 (Section 4.1); Exhibit 2; Appendix A	10
Q2	2. The RFI states that AI use in clinical care poses new challenges for policy and regulation, and that HHS seeks input on potential policy changes to safely accelerate adoption. What specific changes, including applicable CFR citations, should HHS consider?	Question 2 (Section 4.2); Exhibit 3; Exhibit 1A; Appendix C; Appendix E	13
Q3	3. The RFI states that the complexity of AI technology raises new challenges for implementation, including risks to privacy, civil rights and civil liberties. What issues should be considered to best address these concerns and which HHS offices should be involved?	Question 3 (Section 4.3); Exhibit 7; Appendix E	16
Q4	4. The RFI states that robust safety and effectiveness monitoring is essential. What evaluation methods and metrics should be used and how should monitoring be conducted?	Question 4 (Section 4.4); Exhibit 4; Appendix B; Appendix C	17
Q5	5. The RFI states that private sector accreditation, certification, and industry-driven testing can promote effective and safe AI use. How can HHS best support private sector activities (e.g., accreditation, certification, industry-driven testing, and credentialing) to promote innovative and effective AI use in clinical care?	Question 5 (Section 4.5); Exhibit 3; Exhibit 4	19
Q6	6. The RFI seeks real-world examples of where AI tools met or exceeded cost and performance expectations, and where they fell short. In addition, are there novel AI tools that have high potential to improve clinical care, but are not currently in widespread use? Please provide examples.	Question 6 (Section 4.6); Exhibits 3–4; Exhibit 7	20
Q7	7. The RFI asks: What are the most significant decision-makers and stakeholders for adoption of AI tools in clinical care, and what are key administrative, workflow, and resource hurdles to adoption?	Question 7 (Section 4.7); Exhibit 5; Exhibit 3; Appendix C	23
Q8	8. The RFI asks: Where would enhanced interoperability widen market opportunities, fuel research, and accelerate the development of AI for clinical care? Please consider specific data types, data standards, and benchmarking tools.	Question 8 (Section 4.8); Exhibit 6; Exhibit 3; Appendix C	24
Q9	9. The RFI asks: What challenges within health care do patients and caregivers wish to see addressed by use of AI? Equally, what concerns do patients and caregivers have about the use of AI in clinical care?	Question 9 (Section 4.9); Exhibit 7; Appendix E	26
Q10	10. Are there specific areas of AI research that HHS should prioritize to accelerate the adoption of AI as part of clinical care?	Question 10 (Section 4.10); Exhibit 8	27
Q10a	a. Are there published findings about the impact of adopted AI tools and their use clinical care?	Question 10a (Section 4.10a); Exhibit 8; Appendix B	28
Q10b	b. How does the literature approach the costs, benefits, and transfers of using AI as part of clinical care?	Question 10b (Section 4.10b); Exhibit 8	28

Exhibit 1A. What HHS should do (summary)

HHS component	Mechanism	Action / deliverable	Time horizon
ASTP/ONC	Guidance / playbook	Publish a risk-tier trigger rubric and minimum governance + auditability expectations for non-device clinical AI / decision support, explicitly scoped to interoperability and accountability practices and not FDA device jurisdiction.	0–6 months
ONC Certification Program	Implementation guidance + test procedure updates	Clarify that DSI-aligned transparency is necessary but not sufficient, and add testable expectations for runtime auditability (version/config, provenance, user action capture/export) when AI materially influences documentation/orders.	6–18 months
ONC / ASTP	Standards implementation guidance	Publish recommended FHIR profiles/patterns for recording AI tool identity, version/configuration, and provenance for AI-influenced notes, orders, and decision support.	6–18 months
OCR	Implementation guidance	Issue guidance on operationalizing 45 CFR § 92.210 for patient care decision support tools, including what constitutes “reasonable efforts” to identify and mitigate discrimination risk and how to document ongoing monitoring and mitigation.	0–6 months
CMS / CMMI	Model participation terms + reporting	Add participation/reporting terms in CMMI models that require deployers to maintain and produce an AI audit artifact package and post-deployment monitoring signals when AI materially influences clinical decisions or documentation.	6–18 months
CMS (QPP/MIPS)	Program scoring pathways (candidate rulemaking / subreg guidance)	Create an Improvement Activity and/or Promoting Interoperability attestation pathway for maintaining governance artifacts and enabling provenance/audit export for AI-influenced documentation/orders.	6–18 months
CMS (Hospital Quality)	Hospital quality reporting guidance	In Hospital IQR program guidance, specify documentation expectations when AI materially influences eligibility/ordering and require retention of governance artifacts sufficient to reconstruct AI-influenced care processes.	6–18 months
AHRQ	Grants / contracts / implementation science	Fund implementation science on human factors failure modes, monitoring dashboards, and “safety signal” operations for AI-influenced workflow (overrides, data quality, disparity signals, incident response).	6–18 months
AHRQ + ONC + NIST (and partners)	Convening + shared test infrastructure	Stand up an open, governance-backed test harness/benchmark suite for common clinical workflow scenarios that includes provenance-aware writeback and auditing patterns, enabling reproducible evaluation and procurement-ready conformance testing.	18+ months

3. Understanding of HHS Context and Objectives: Adoption depends on governance, interoperability, and accountability in real workflows

This RFI appropriately frames clinical AI adoption as a multi-component HHS problem that spans interoperability, safety, privacy, civil rights, reimbursement incentives, and evidence generation.

We interpret HHS's objective as enabling clinically meaningful use of AI with accountable oversight, without creating duplicative or conflicting regulatory regimes.

We also note that a substantial computable guideline, eCQM, and clinical decision support ecosystem already exists (e.g., eCQMs, CDS artifacts, SMART-based apps), and our recommendations are intended to be complementary by improving auditability, provenance, verification of internal logic properties where appropriate, and workflow integration.

Evaluator-relevant objectives reflected in this comment

- Differentiate AI technologies by runtime behavior and risk profile (e.g., deterministic decision artifacts vs probabilistic inference services) to apply proportionate controls.
- Emphasize testability, provenance, and retrospective review for higher-risk non-device AI used in clinical workflows.
- Reduce adoption friction through standards-based interoperability patterns (structured data capture, app launch, event-driven CDS, and provenance-aware writeback).
- Use federal payment and program mechanisms to shift demand toward tools that are measurable, governable, and aligned to guideline-concordant care.
- Support learning health system approaches through reusable evaluation materials (test cases, reporting templates, and monitoring metrics) that can be applied across sites.

Exhibit 2 summarizes why deterministic compiled artifacts and probabilistic AI services warrant different evaluation and monitoring expectations.

Exhibit 2. Deterministic compiled artifacts and probabilistic AI services warrant different controls

Dimension	Deterministic compiled artifact	Probabilistic AI service
Runtime behavior	Deterministic execution; no inference-time probabilistic reasoning.	Runtime inference; outputs may vary and depend on model/prompt/version.
Primary safety risk	Logic errors/omissions and update governance gaps.	Stochastic behavior, drift, and prompt/model changes.
Evaluation emphasis	Build-time verification and traceability to guideline version/text.	Pre/post deployment performance and robustness; subgroup analysis.
Monitoring emphasis	Version control, regression testing, provenance-aware audit artifacts.	Operational monitoring, thresholds, escalation, incident response.
Evidence basis for recourse	Guideline/page/paragraph and encoded logic path.	Reconstruction from logs (inputs, model/version, prompts/config, user actions).

Operational risk-tier triggers (qualitative rubric)

- Does the output influence diagnosis, treatment, medication, triage, eligibility, or access to services (vs purely administrative support)?
- Is the output patient-facing (directly consumed by patients/caregivers) or clinician-facing only?
- Is the tool autonomous (acts by default) or clinician-confirmed (requires deliberate acceptance/attestation)?
- Does the tool generate or modify the legal medical record (documentation/writeback) or only provide a transient suggestion?
- What is the scale of use (single clinic vs enterprise vs multi-tenant service) and the speed of propagation of errors?
- How reversible is harm (can incorrect outputs be quickly detected and corrected before patient impact)?
- Does the tool use protected class variables or proxies in ways that raise discrimination risk (triggering heightened nondiscrimination governance)?

Operational risk-tier triggers (qualitative rubric)

- **Tier 1 (low):** administrative or informational support; clinician review; no autonomous actions; minimal audit artifacts and change logs.
- **Tier 2 (moderate):** influences documentation/orders/clinical decisions but remains clinician-confirmed; full “AI audit artifact package” plus monitoring of overrides, data quality, and disparate application.
- **Tier 3 (higher):** patient-facing and/or affects triage/access and/or operates with default/autonomous actions at scale; full audit artifact package plus enhanced evaluation, incident response, and nondiscrimination documentation aligned to 45 CFR § 92.210.

Exhibit 2A. HHS objectives and jurisdiction-aware framing

HHS objective	Why it matters for adoption	Implementation-oriented framing
Enable accountable adoption of clinical AI across settings and vendors	Adoption stalls when decision-makers cannot verify what a tool did, under which configuration, and how it affected documentation/orders.	Publish risk-tiered minimum governance + auditability expectations for non-device clinical AI/decision support, tailored by runtime behavior and intended use.
Improve interoperability for AI-enabled workflow integration	AI that cannot integrate with EHR workflows and standards-based writeback creates operational risk and limits measurable benefit.	Use ONC Certification implementation guidance and testing to require standards-based capture/export of provenance and user actions for AI-influenced documentation/orders (SMART + CDS Hooks + SDC + Provenance).
Support privacy and security without over-prescribing implementation	Clinicians and organizations need confidence that AI does not introduce uncontrolled data flows and that audit controls exist for access and modification events.	Reference HIPAA Security Rule audit controls and require documented retention/access policies for AI-related logs and artifacts rather than implying fixed retention durations.
Uphold civil rights and nondiscrimination	Patient care decision support can amplify inequities if inputs/outputs are not monitored and mitigations are not documented.	Operationalize OCR's 45 CFR § 92.210 through guidance that ties "reasonable efforts" to concrete governance artifacts, monitoring, and mitigation workflows.
Create aligned incentives (avoid pure fee-for-service distortions)	If governance and monitoring are unfunded "uncompensated compliance," adoption will skew toward well-resourced systems and vendors.	Use CMS program terms and quality pathways to incent governance artifacts and monitoring proportional to risk, without defaulting to new standalone "AI billing codes."

4. Responses to RFI Questions (in RFI order)

The sections below respond to each RFI question in the order presented, retaining the verbatim question text for evaluator convenience.

Recommendations are scoped to HHS-implementable levers (guidance, certification implementation/testing, program terms, quality pathways, grants/contracts) and are tailored by runtime behavior and risk tier.

Question 1: Adoption barriers are governance, interoperability, and operational accountability

RFI question: 1. What are the biggest barriers to private sector innovation in AI for health care and its adoption and use in clinical care?

In practice, the most material barriers to innovation and adoption often emerge after development. The friction typically resides in governance, interoperability, and operational accountability once AI outputs can influence clinical decisions, documentation, or access to services.

Key barriers observed across deployments

- **Lack of a regulatory distinction between deterministic and probabilistic clinical AI:**Current frameworks often treat all AI in clinical care as a single category, applying similar evaluation, monitoring, and governance expectations whether a tool performs probabilistic runtime inference or executes a pre-compiled, formally verified decision artifact. This can overburden deterministic tools with requirements such as drift monitoring for logic that cannot drift, while underspecifying oversight for non-deterministic systems whose runtime behavior is inherently less predictable.
- **Accountability and auditability gaps:**Many deployments cannot reliably support retrospective reconstruction of what the tool recommended, under which version and configuration, what inputs were used, and how the output affected documentation, orders, task routing, or clinician actions. This limits incident investigation, documentation integrity review, and organizational trust.
- **Workflow integration and interoperability burden:**AI tools often fail to produce measurable clinical benefit when they are not integrated into ordering, documentation, and task routing workflows. Common barriers include launch context, authorization, latency, writeback, provenance capture, and variability across certified health IT implementations. Inconsistent support for standards-based invocation and data exchange makes implementation costly and brittle across sites.
- **Unstructured guidance and inconsistent clinical data capture:**Clinical guidance is often ambiguous or not computable, and data capture practices vary substantially across organizations. This reduces reproducibility, increases site-specific customization, and complicates cross-site evaluation.
- **Change-control and update management gaps:**Organizations frequently lack practical processes for safe updates, including regression testing, version transition controls, rollback procedures, and governance over rapid tool iterations and guideline changes. These gaps create operational risk and slow deployment.

- **Monitoring gaps not tailored to risk and runtime behavior:** Many deployments lack feasible post-deployment monitoring that is proportional to risk and appropriate to runtime behavior. Deterministic and probabilistic tools require different monitoring approaches, but deployers often lack clear operational guidance on what is sufficient and practical.
- **Unclear minimum governance expectations for non-device AI:** For non-device AI used in clinical care, there is often no clear baseline for auditability, provenance, update control, performance monitoring, and recourse. This uncertainty slows procurement and results in inconsistent controls across organizations.
- **Procurement, legal, and contracting friction:** Health systems often repeat security, privacy, and clinical governance reviews across vendors and products. Adoption is further slowed by unclear allocation of responsibility between developers and deployers for tool errors, documentation integrity, and nondiscrimination obligations.
- **Equity and civil rights implementation complexity:** Deployers often lack practical guidance on how to document and operationalize reasonable efforts to identify and mitigate discrimination risks, especially when protected characteristics may be proxied indirectly.
- **Resource constraints for smaller, rural, and safety-net providers:** Many providers cannot absorb the ongoing costs of governance, integration, evaluation, and monitoring without aligned incentives, shared infrastructure, or technical assistance. This risks widening inequities in adoption and access.
- **Limited reusable evaluation and conformance tools:** There is limited availability of standardized test cases, reporting templates, and post-deployment monitoring metrics that can be reused across sites. As a result, organizations duplicate validation and integration work.

Opportunities for Acceleration

- **Standardize a portable minimum audit evidence package:** HHS can define a baseline AI audit artifact package that is reusable across products and sites, including intended use, versioning, provenance, configuration, validation summary, monitoring plan, and recourse process. This would reduce duplicative review burden and improve accountability.
- **Build on existing ONC transparency and intervention risk foundations:** HHS can extend existing ONC transparency and intervention risk approaches toward runtime traceability that supports audits, incident response, and continuous improvement in clinical operations.
- **Strengthen existing computable clinical infrastructure:** HHS can accelerate adoption by improving current ecosystems, including computable guidelines, CDS artifacts, eQMs, SMART apps, and standards-based workflows, through stronger provenance, versioning, verification, and conformance practices rather than replacing them.
- **Create shared, governance-backed test harnesses for common workflows:** A shared conformance and testing infrastructure for common integration patterns, including standards-based launch and invocation, structured data capture, and provenance-aware writeback, would reduce custom integration burden and improve procurement readiness.

- **Align payment and program incentives with governable deployment:** Federal payment and program levers can encourage adoption of tools that are auditable, monitorable, and operationally accountable, rather than tools selected primarily on claimed functionality.

Recommendations to HHS

- **Publish risk-tiered minimum governance expectations for non-device AI used in clinical care:** HHS should publish clear, risk-tiered minimum expectations for auditability and operational governance, including versioning, provenance, input/output traceability, update control, monitoring, and recourse, with greater requirements for higher-risk uses.
Why this helps: A predictable baseline reduces procurement uncertainty, supports consistent controls across settings, and improves investigation when safety, equity, or documentation integrity concerns arise.
- **Distinguish deterministic AI-compiled artifacts from probabilistic AI services:** HHS should publish guidance that differentiates AI-compiled deterministic artifacts from probabilistic runtime AI services and sets proportionate evaluation and monitoring expectations for each.
Why this helps: This avoids imposing unnecessary monitoring burdens on deterministic artifacts while ensuring probabilistic systems receive oversight appropriate to their risk.
- **Fund open conformance tools and reference implementations for common integration patterns:** HHS should support reusable conformance tools and reference implementations for standards-based launch, invocation, structured data capture, and provenance-aware writeback.
Why this helps: Interoperability friction is a major source of cost and delay. Shared infrastructure reduces duplicative implementation work and supports safer deployments.
- **Use payment and program policy to require a basic governance package for higher-risk AI-enabled workflows:** For higher-risk AI-enabled workflows, HHS should condition federal payment support or program participation on submission of a basic governance package, such as intended use, monitoring plan, audit artifact package, and recourse process, rather than relying only on claimed functionality.
Why this helps: Payment levers can shift demand toward tools that are governable and measurable and reduce incentives to deploy poorly monitored systems.
- **Provide practical implementation templates for equity and discrimination risk mitigation:** HHS should publish practical templates and examples that help deployers document reasonable efforts to identify and mitigate discrimination risks in implementation and monitoring.
Why this helps: Many organizations support equity safeguards but lack operational clarity. Templates can improve consistency and reduce implementation uncertainty.
- **Support shared infrastructure and technical assistance for smaller and safety-net providers:** HHS should support shared governance resources, reusable evaluation tools, and technical assistance pathways for smaller, rural, and safety-net providers.
Why this helps: This reduces structural barriers to adoption and helps ensure that benefits from AI-enabled care are not limited to large, well-resourced systems.

Question 2: Policy changes should focus on auditability, provenance, and verification, supported by program incentives and CFR anchors

RFI question: 2. What regulatory, payment policy, or programmatic design changes should HHS prioritize to incentivize the effective use of AI in clinical care and why? What HHS regulations, policies, or programs could be revisited to augment your ability to develop or use AI in clinical care? Please provide specific changes and applicable Code of Federal Regulations citations.

HHS should implement a jurisdiction-aware approach that focuses on minimum governance, auditability, and interoperability expectations for non-device clinical AI / decision support, tailored by runtime behavior and clinical risk.

This approach should be explicitly scoped so that it does not alter FDA jurisdiction over medical devices and does not purport to “approve” model quality, but instead makes accountability and reconstruction feasible when AI materially influences clinical care, documentation, or ordering.

Specific changes HHS should consider (with candidate CFR anchors where applicable)

1. **ASTP/ONC:** publish risk-tiered minimum expectations for non-device clinical AI / decision support
 - a. Mechanism: ASTP/ONC guidance/playbook (non-binding) with templates.
 - b. Deliverable: A risk-tier trigger rubric and minimum expectations for governance artifacts, auditability, and monitoring tailored by runtime behavior (deterministic guideline-logic artifacts vs probabilistic inference services).
 - c. CFR anchor concept (ONC program context): ONC Certification Program decision support criteria context at 45 CFR 170.315(b)(11) (DSI).
2. **ONC Certification Program:** make runtime auditability testable where AI materially influences documentation/orders
 - a. Mechanism: Implementation guidance + test procedure updates (and, if needed, future certification criterion refinement through rulemaking).
 - b. Deliverable: Testable capabilities for (a) capturing tool identity + version/configuration at runtime, (b) recording provenance for AI-influenced notes/orders, and (c) exporting relevant audit/provenance data for governance review.
 - c. CFR anchor: 45 CFR 170.315(b)(11) and related ONC certification provisions (see Appendix E for candidate anchors).
 - d. Alignment note: The ONC DSI Resource Guide emphasizes transparency and risk management and clarifies that the program does not “approve” predictive DSI quality; our proposal extends DSI by focusing on runtime auditability that enables retrospective reconstruction.

3. **OCR:** operationalize nondiscrimination expectations for patient care decision support tools
 - a. Mechanism: OCR implementation guidance and examples.
 - b. Deliverable: Practical documentation and monitoring guidance for “reasonable efforts” to identify and mitigate discrimination risk when using patient care decision support tools, especially where protected class variables are used or proxied.
 - c. CFR anchor: 45 CFR § 92.210 (patient care decision support tools) and related Part 92 requirements (see Appendix E).
4. **CMS:** incent governance artifacts and monitoring through existing program levers rather than new standalone “AI billing codes”
 - a. Mechanism (three implementation routes):
 - i. CMMI model participation/reporting terms (model-specific terms and reporting requirements)
 - ii. MIPS (QPP) Improvement Activities and Promoting Interoperability pathways
 - iii. Hospital quality reporting guidance (e.g., Hospital IQR) where AI materially influences eligibility/ordering/documentation
 - b. Deliverable: Program terms and attestations that require maintenance of an AI audit artifact package and tailored post-deployment monitoring signals when AI materially influences care processes.
 - c. Candidate CFR anchors for counsel validation: 42 CFR Part 414, Subpart O (Quality Payment Program), including 42 CFR § 414.1375 (Promoting Interoperability) and related Improvement Activities provisions, and 42 CFR § 412.140 (Hospital IQR Program).

Reimbursement realism (standalone billing vs incentives)

The RFI correctly highlights that fee-for-service incentives can distort adoption toward volume rather than accountability.

We recommend CMS avoid creating a standalone “AI billing code” as the default policy response and instead tie incentives to (1) outcomes/value-based constructs, (2) validated burden reduction in bounded workflows, and (3) auditable governance and evaluation artifacts rather than model claims.

This framing supports adoption while reducing incentives for overuse, underdocumentation of risks, or deployment without monitoring.

Exhibit 3 defines the minimum “AI audit artifact package,” and Appendix C provides an expanded version intended for procurement and governance use.

Exhibit 3. Recommended minimum AI audit artifact package (field set)

Minimum artifact element	Purpose (what it enables)
Tool identity and scope (name, developer, intended use, clinical context, contraindicated uses)	Prevents repurposing beyond intended use and supports appropriate governance review.
Version and configuration at runtime (model/artifact version, prompt/config parameters where applicable, enabled features)	Supports reconstruction of what actually ran for a given output and enables safe update governance.
Input data dependencies (data types used, required fields, known missingness sensitivities)	Supports data quality monitoring and explains failure modes tied to missing/incorrect inputs.
Output characterization (output type, confidence/uncertainty handling where applicable, recommended user action)	Clarifies how outputs should be interpreted and used in workflow.
Known limitations and failure modes (including fairness/robustness considerations)	Supports safe use, informed oversight, and targeted monitoring.
Provenance + writeback behavior (what is written to the record, where, and how provenance is attached)	Protects record integrity and enables traceability for downstream consumers and audits.
User interaction capture (accept/override/modify, escalation paths)	Enables monitoring of reliance, overrides, and human-in-the-loop performance.
Update/change log and regression testing evidence (including guideline/version changes for deterministic artifacts)	Enables controlled updates and reduces the risk of silent behavior changes.

Exhibit 3A. Clarifying “who must produce which artifacts” (implementability)

Actor	Primary responsibility	Examples of artifacts/capabilities
Certified health IT / EHR module	Capture and export runtime evidence of AI use in workflow.	Provenance attachment to writeback targets; exportable user action logs; standardized interfaces (SMART/CDS Hooks/FHIR) for invocation and data access; audit event export patterns.
AI developer (tool supplier)	Provide tool identity, limitations, versioning, and change transparency.	Version/config disclosure; known limitations; change logs; intended use; interface specifications; safety and fairness limitations statements.
Deployer (health system/clinic)	Define local intended use, governance, monitoring, and recourse.	Local configuration documentation; monitoring dashboards and review cadence; incident response and escalation; patient/clinician recourse processes; nondiscrimination documentation aligned to 45 CFR § 92.210 where applicable.

Question 3: Novel issues are practical: allocation of responsibility, update governance, record integrity, and audit access

RFI question: 3. For non-medical devices, we understand that use of AI in clinical care may raise novel legal and implementation issues that challenge existing governance and accountability structures... What novel legal and implementation issues exist and what role, if any, should HHS play to help address them?

For non-device AI that influences clinical decisions, documentation, or access decisions, the most consequential legal and implementation issues are often operational: reproducibility, allocation of responsibilities, update governance, and record integrity.

Issues that commonly create compliance and safety risk

- **Privacy and security (HIPAA and beyond):** AI integrations increase data flows and new access pathways, so deployers need audit controls and documented retention/access policies appropriate to their risk tier and workflow.
- **Record integrity and provenance:** when AI generates or influences documentation/orders, organizations must be able to identify AI-influenced content, reconstruct inputs/outputs, and preserve clinician accountability for final actions.
- **Civil rights and nondiscrimination:** patient care decision support tools can introduce discriminatory effects through input selection, proxies, workflow routing, or differential error rates across pop.
- **Patient trust and autonomy:** patient-facing or patient-impacting AI requires clear disclosures, meaningful recourse, and safeguards against deceptive or coercive automation.
- **Liability and responsibility allocation:** contracts and governance should allocate responsibility for tool defects, configuration errors, data quality failures, and monitoring omissions.
- **Information access and information blocking:** interoperability obligations must be implemented carefully within existing exceptions frameworks rather than implying blanket access rights.
- **Patient rights applicability:** access/amendment rights depend on whether information is maintained in the designated record set, and disclosures should avoid overpromising rights that are context-dependent.

Which HHS offices should be involved (and why)

- **ASTP/ONC:** to lead interoperability and certification-aligned auditability expectations and convene standards implementation guidance.
- **OCR:** to provide implementation guidance for nondiscrimination obligations for patient care decision support tools under 45 CFR § 92.210 and for privacy-related considerations in patient-facing tools.
- **CMS/CMMI:** to align incentives and reporting expectations (governance artifacts and monitoring) through program terms and quality pathways.
- **AHRQ:** to fund implementation science on human factors, monitoring operations, and equitable deployment practices, and to support evaluation infrastructure.
- **OIG (as appropriate):** to align compliance program guidance and oversight concepts with auditability expectations, especially where AI affects documentation, coding, or medical management processes.

Question 4: Evaluation should be staged and risk-tiered, with formal verification for deterministic guideline logic

RFI question: 4. For non-medical devices, what are the most promising AI evaluation methods (pre- and post-deployment), metrics, robustness testing, and other workflow and human-centered evaluation methods for clinical care? Should HHS further support these processes? If so, which mechanisms would be most impactful (e.g., contracts, grants, cooperative agreements, and/or prize competitions)?

Evaluation should be proportional to risk tier and tailored to runtime behavior, with emphasis on reproducible measurement and feasible post-deployment monitoring.

HHS should avoid implying that any single evaluation method “proves” clinical correctness, and should instead emphasize traceability, auditability, and post-deployment safety monitoring.

Deterministic guideline-logic artifacts can support formal verification of internal logical properties (e.g., consistency, reachability, completeness over defined states), but this does not prove clinical benefit, data quality, or appropriate interpretation across contexts.

Exhibit 4 summarizes evaluation components for higher-risk non-device clinical AI across pre- and post-deployment stages, including human-centered and equity-focused methods.

Exhibit 4. Risk-tiered evaluation lifecycle for non-device clinical AI

Lifecycle stage	Minimum evaluation elements (all tools)	Tailoring by runtime behavior
Pre-deployment (fit-for-use)	Define intended use, workflow, and user role; identify failure modes; validate data availability/quality; perform baseline performance testing in representative settings.	Deterministic: verify internal logical properties over defined states and perform regression tests against test cases for guideline/version changes. Probabilistic: evaluate discrimination risk, calibration/uncertainty handling where applicable, robustness to distribution shifts, and performance across subgroups.
Implementation (integration + governance)	Document tool identity/version/config; define human-in-the-loop responsibilities; establish incident response and escalation; ensure provenance + user action capture.	Deterministic: ensure executable logic is version-controlled and linked to source guideline versions and local policy choices. Probabilistic: ensure model update policy is explicit and changes are auditable.
Post-deployment monitoring (safety signals)	Monitor for safety signals proportional to risk tier; review overrides/escalations; monitor data quality/missingness; monitor disparate application/outcomes; maintain recourse pathways.	Deterministic: no statistical drift monitoring, but monitor data quality, overrides/escalations, disparate application, and guideline/version update impacts with regression test outcomes. Probabilistic: include performance monitoring and drift detection where appropriate in addition to the above.
Periodic re-evaluation (learning + accountability)	Periodic governance review, documentation updates, and evaluation refresh based on incidents, updates, or workflow changes.	Deterministic: re-run regression tests when guidelines, local policies, or input mappings change. Probabilistic: re-evaluate when model retrains, input features change, or drift signals occur.

Transparent reporting using established guidance (e.g., CONSORT-AI, SPIRIT-AI, and DECIDE-AI) can support comparability across tools and contexts when supplemented with deployment context and governance artifacts for non-device AI.

Formal verification for deterministic clinical decision logic

For clinical AI that encodes guideline logic as deterministic decision trees (as opposed to runtime probabilistic inference), evaluation can include formal verification of logical properties using established methods from safety-critical software engineering.

Satisfiability Modulo Theory (SMT) solvers can prove, across the full input space rather than a finite test suite, that: (a) every possible combination of patient inputs leads to at least one recommendation (exhaustiveness); (b) no combination triggers contradictory recommendations (determinism); and (c) every encoded recommendation is reachable by at least one patient profile (no dead rules).

Graph-theoretic analysis can verify structural properties including acyclicity (no infinite assessment loops), connectivity (no orphan recommendations), and minimum path length (minimum patient burden).

These methods are established practice in aviation software and nuclear safety systems. They are directly applicable to clinical decision logic encoded in standards-based formats such as FHIR Structured Data Capture. HHS should fund development of formal verification tooling and reference implementations for clinical decision artifacts, and should recognize formal verification evidence in evaluation and procurement frameworks.

Transparency is necessary but insufficient for clinical decision logic that determines screening eligibility and treatment pathways; the logical correctness of the encoded artifact should be provable, even though clinical benefit and safe use in context must still be demonstrated through empirical evaluation and post-deployment monitoring.

Recommendations to HHS

- **R&D:** Support development and adoption of standardized evaluation and reporting templates for non-device clinical AI that can be completed without disclosing proprietary model details, aligned to CONSORT-AI, SPIRIT-AI, and DECIDE-AI.
Why it helps: Standard templates reduce duplicative procurement burden and improve comparability across sites and tools, accelerating a learning health system approach.
- **R&D:** Use grants, cooperative agreements, and prize competitions to fund pragmatic, multi-site evaluations that measure workflow impact, safety outcomes, and equity, and that require version and provenance reporting.
Why it helps: Real-world performance depends on local context; pragmatic evaluation reduces the gap between retrospective technical performance and deployed impact.
- **Regulation:** Encourage risk-tiered post-deployment monitoring expectations (metrics, thresholds, and escalation) anchored to NIST AI RMF functions (GOVERN, MAP, MEASURE, MANAGE) and compatible with DSI intervention risk management concepts.
Why it helps: A governance anchor helps operationalize monitoring and response without prescribing a single technical method, enabling proportionate control across AI types and risks.

Question 5: Accreditation works when it certifies what is testable: conformance, auditability, and monitoring readiness

RFI question: 5. How can HHS best support private sector activities (e.g., accreditation, certification, industry-driven testing, and credentialing) to promote innovative and effective AI use in clinical care?

HHS can support private sector accreditation and testing by defining a clear “floor” of auditable governance expectations and enabling conformance testing that does not require HHS to certify model quality.

A practical approach is to support accreditation/testing programs that verify the presence and exportability of governance artifacts (identity/version/config, provenance, user action capture, monitoring plans) and the ability to run reproducible workflow tests.

Alignment with ONC DSI (reduce duplication and increase credibility)

ONC’s DSI framework already emphasizes transparency and intervention risk management and explicitly does not aim to approve or guarantee predictive DSI quality.

Our proposal extends (not replaces) DSI “source attributes” and intervention risk management concepts by focusing on runtime auditability—version/config traceability, provenance, and user actions—that enables retrospective reconstruction and supports monitoring and incident response.

Concrete HHS support actions

- **RASTP/ONC:** convene a public-private technical working group to define minimal conformance tests for AI audit artifacts and provenance-aware writeback in common workflows.
- **AHRQ:** fund and evaluate independent testing protocols and implementation science studies of accredited programs’ impact on safety, equity, and operational burden.
- **CMS/CMMI:** recognize accredited governance/auditability conformance as evidence in program participation/reporting terms, without treating accreditation as proof of clinical benefit.
- **OCR:** provide guidance to ensure accreditation programs incorporate nondiscrimination documentation consistent with 45 CFR § 92.210 when patient care decision support tools are in scope.

Question 6: AI value is most consistent in tight, reviewable workflows; verified preventive-care logic remains high potential

RFI question: 6. Where have AI tools deployed in clinical care met or exceeded performance and cost expectations and where have they fallen short? What kinds of novel AI tools would have the greatest potential to improve health care outcomes, give new insights on quality, and help reduce costs?

The clearest operational wins so far have come from narrow, high-volume workflows where outputs are reviewable and integration is tight. AI tools are most likely to meet or exceed expectations when intended use is clear, workflow boundaries are defined, and organizations can monitor how outputs are actually used.

AI has fallen short when inputs are poorly structured, workflow integration is weak, governance artifacts are incomplete, or updates are pushed without clear change control. Those gaps create safety and compliance risk and make it harder to tell whether a tool is creating real value or just shifting work and cost elsewhere.

Below are short, de-identified vignettes that reflect common deployment patterns, with lessons tied to governance, auditability, and interoperability:

- **Ambient documentation (met expectations in bounded workflows):** In multiple deployments, ambient documentation tools have reduced clinician documentation burden and improved perceived usability when deployed with clear intended use and workflow boundaries.

Where these deployments perform best, organizations can reconstruct tool version/configuration and can review edits/overrides, enabling safe iteration and targeted monitoring.

Lesson: success depends on provenance-aware writeback and user action capture more than on headline accuracy claims, because the record integrity and clinician accountability requirements are operationally determinative.

- **Imaging worklist triage (met expectations when integrated and monitored):** In settings using imaging triage/priority tools, value is most evident when integration supports measurable time-to-review improvements and when outputs are monitored for subgroup performance and alerting fatigue.

Where these tools fall short, the common failure mode is not a single “bad model,” but poor integration into clinical routing and insufficient auditing of when/why clinicians override.

Lesson: auditability and monitoring should be treated as part of the intervention, not optional post-hoc documentation.

- **Patient-facing symptom checkers (often fell short on diagnostic accuracy and consistency):**

Systematic reviews continue to find low and variable diagnostic accuracy across online symptom checkers, with triage accuracy generally higher but still variable across tools and studies.

Common failure modes include incomplete data capture, inconsistent recommendations across similar presentations, limited transparency about limitations, and unclear recourse when outputs are wrong.

Lesson: patient-facing or patient-impacting tools should trigger higher risk-tier expectations, including clear disclosures, documentation of limitations, and ongoing monitoring and mitigation, aligned to nondiscrimination duties where applicable.

- **Administrative utilization/authorization support (mixed results with transfer/payment effects):** Tools that support prior authorization workflows can reduce staff burden or speed decisions in narrow use cases, but may also shift costs and burdens across payers, providers, and patients depending on how criteria are implemented and how appeals/overrides are handled.

Where deployments fall short, it is often due to opaque criteria application, limited audit trails for why an outcome occurred, and insufficient recourse for correction.

Lesson: governance artifacts and traceability are necessary to manage transfer effects and to support fair, appealable, and auditable processes.

Promising areas for novel tools (as observed)

- **Preventive care identification and outreach grounded in computable guideline logic:** A high-potential category is preventive care identification and outreach built on structured data capture and published guidelines, with traceable logic and auditable outputs.

One promising approach is AI-assisted compilation of guideline artifacts into constrained, reviewable workflow logic (including deterministic decision trees where appropriate), paired with clinician review and interoperable writeback. This can support point-of-care identification of eligible services, more consistent outreach, and clearer documentation of why a recommendation was made.

This category remains especially promising because the core problem is often not lack of evidence-based recommendations, but unreliable translation of those recommendations into repeatable workflow execution.

- **Care coordination and navigation tools with provenance-aware workflows:** strong potential when they reduce friction across settings while preserving provenance, role-based access control, and reviewable workflow actions.

High-value use cases include referral completion, follow-up scheduling, preventive service completion, and transitions across care settings. The opportunity is not only automation of tasks, but better continuity and accountability across fragmented systems.

- **Decision support that writes back reviewable, provenance-tagged artifacts:** could improve quality and efficiency if outputs are clearly reviewable and correctable before they affect the record or downstream decisions.

The design requirement is critical: AI-generated suggestions should remain distinguishable from clinician-authored content, with clear attribution and user action capture.

- **Risk-tiered operational monitoring and “safety signal” dashboards** have substantial potential to make AI deployment performance visible in routine operations, rather than only during periodic review.

Examples include dashboards that track override rates, missingness/data quality patterns, subgroup performance signals, incident reports, alert fatigue, and workflow delays. These tools could improve early detection of unintended consequences and support safer scaling, procurement, and governance.

- **Documentation and coding support that preserves record integrity:** continue to improve administrative efficiency, but the highest-value designs will preserve record integrity through provenance, attribution, and separation of suggestions from finalized clinician documentation.

This is especially important where generated content may affect billing, quality measurement, audits, or downstream clinical decision-making. Efficiency gains are more durable when organizations can reconstruct what was suggested, what was accepted, and what was changed.

Evidence referenced on preventive care screening gaps

Analysis of 227 million T-MSIS Medicaid claims rows, cross-validated against 2.99 million CDC BRFSS survey responses and HRSA UDS data from 18,751 FQHC sites, identifies a confounder-adjusted \$320-450 million annual Medicaid billing gap in USPSTF-recommended preventive services.

Self-reported screening rates from BRFSS confirm the gap is real and worsening: national breast cancer screening completion declined from 77.7% in 2014 to 74.4% in 2024. Seventy-seven percent of FQHC sites report below the 75th percentile on three or more HRSA-mandated screening measures.

This gap is not attributed in this submission to coverage barriers. ACA Section 2713 mandates \$0 copay for USPSTF A and B grade services, affecting 150-180 million insured Americans. The gap is attributed here to the absence of computable, verifiable clinical decision logic at the point of encounter.

AI-compiled guideline artifacts that encode USPSTF recommendations as formally verified decision trees (deterministic, zero marginal cost per patient, FHIR-compliant) are presented as one approach to address this supply-side bottleneck directly.

Question 7: Adoption decisions span clinical, IT, privacy, and finance; standard checklists can reduce administrative burden

RFI question: 7. Which role(s), decision maker(s), or governing bodies within health care organizations have the most influence on the adoption of AI for clinical care? What are the primary administrative hurdles to the adoption of AI in clinical care?

Adoption decisions for clinical AI are multi-stakeholder decisions, and adoption often fails when governance artifacts and monitoring responsibilities are not clearly assigned. Decision-makers require confidence that a deployment can be audited, monitored, and updated safely, and that responsibilities are clear across developer, EHR, and deployer roles.

Exhibit 5 summarizes the decision makers who commonly influence adoption decisions and the recurring administrative hurdles that slow deployment.

Exhibit 5. Typical decision makers and administrative hurdles for clinical AI adoption

Decision-maker / stakeholder	Adoption questions each must answer	Primary administrative hurdles
Clinical leadership (CMIO/CMO, nursing leadership, service line leads)	Does the tool fit clinical workflow, preserve clinician accountability, and have clear intended use/limitations and escalation pathways?	a. Duplicated and inconsistent security/privacy reviews (including uncertainty about log retention and access controls) b. Integration complexity and variability across EHR implementations (launch, authorization, writeback) c. Limited standardized documentation for evaluation and monitoring d. Update management and change control concerns e. Lack of category-specific evaluation criteria (deterministic vs probabilistic), driving duplicative and inconsistent reviews
Quality and safety leadership (quality, patient safety, risk management)	Can we monitor safety signals, reconstruct events, and manage incidents, including disparate impact and recourse?	
IT leadership (CIO/CTO, integration teams)	Can we integrate via standards-based interfaces (SMART/CDS Hooks/FHIR), support provenance-aware writeback, and avoid brittle custom interfaces?	
Security and privacy (CISO, HIPAA security, privacy officer)	Are audit controls in place, and do we have documented retention/access policies and risk analysis appropriate to the data flows and risk tier?	
Compliance and legal (including civil rights compliance)	Are nondiscrimination obligations addressed (45 CFR § 92.210 where applicable), and are contracting, liability, and documentation integrity responsibilities clear?	
Operations and finance (administration, revenue cycle where relevant)	Are benefits measurable and sustainable, and are costs and transfer effects (who pays vs who benefits) understood and managed?	

Key hurdles

- Operational burden of governance and monitoring without standardized, exportable artifacts.
- Integration costs and variability across EHRs and sites when standards-based patterns are not consistently implemented.
- Unclear update governance (includes regression testing) and inability to manage rapid iteration safety management
- Unclear accountability for record content when AI-generated content is written back without provenance.
- Limited staff and expertise to run ongoing monitoring, especially in smaller and safety-net settings.

Question 8: Interoperability priorities are structured capture, standard launch/invocation, and provenance-aware writeback

RFI question: 8. Where would enhanced interoperability widen market opportunities, fuel research, and accelerate the development of AI for clinical care? Please consider specific data types, data standards, and benchmarking tools.

Enhanced interoperability widens market opportunities when it reduces per-site integration costs, enables portable governance artifacts, and supports reproducible evaluation across settings.

Interoperability should be framed not only as “data access,” but as end-to-end workflow integration: invocation, context passing, writeback, and provenance/audit export that supports governance and research.

Specific data types and standards (examples, non-exhaustive)

- Core clinical data via USCDI/US Core (problems, meds, allergies, labs, vitals, notes, procedures, immunizations) to support broad clinical AI inputs and evaluation.
- Orders, results, and documentation writeback targets with provenance for AI-influenced ordering and note generation.
- Quality measurement data pathways (eCQMs and related constructs) to support evaluation and monitoring aligned to existing measurement ecosystems.
- Patient-generated and patient-facing data pathways where relevant, with heightened privacy and nondiscrimination governance.

Benchmarking tools (what is most useful for HHS to support)

Benchmarking that matters for clinical care is workflow-centric and provenance-aware, not only model-centric.

HHS should support benchmark suites and test harnesses that evaluate integration patterns (SMART launch context, CDS Hooks invocation, provenance-aware writeback, audit export) in representative scenarios, enabling reproducible evaluation and procurement-ready conformance testing.

Exhibit 6 summarizes interoperability priorities and why they matter for scalable, governable clinical AI deployments.

Exhibit 6. Interoperability building blocks for scalable clinical AI

Building block	Role in clinical AI integration	Why it matters for governance and auditability
SMART App Launch (FHIR) + scopes	Standards-based app launch, context passing, and authorization for data access.	Enables consistent access control and supports auditable, least-privilege integration patterns.
CDS Hooks	Event-driven invocation for decision support at workflow moments (ordering, chart open, etc.).	Supports traceable “when invoked” events and structured context exchange for auditing.
FHIR Provenance (and related audit patterns)	Represent who/what generated data and how it was derived, including AI-influenced content.	Enables identification of AI-influenced record content and supports retrospective reconstruction.
FHIR Structured Data Capture (SDC)	Structured capture, prepopulation, and extraction support for forms and questionnaires.	Supports consistent input capture and downstream extraction, but is not the sole reasoning substrate for deterministic guideline logic.
Computable guideline / CDS representations (multiple formats)	Represent deterministic eligibility logic and guideline-driven decision support in computable forms.	Supports auditability and regression testing when linked to provenance, versioning, and source guideline references.
Callable tool/API invocation interfaces (optional architecture note)	A generic pattern for invoking external tools/services in a controlled way.	Useful as an implementation pattern, but not a core HHS policy recommendation; auditability expectations should be architecture-neutral.

Based on these building blocks, HHS can improve comparability across deployments by encouraging a small set of minimum conformance expectations for interoperable clinical AI integrations.

Minimum conformance expectations

- Implement standards-based invocation (for example, SMART and/or CDS Hooks) with documented context passing.
- Maintain provenance-aware writeback for AI-influenced documentation or orders where writeback occurs.
- Capture/export user interaction signals (accept, override, modify) for governance monitoring.
- Capture/export runtime tool metadata (identity, version, configuration) for retrospective reconstruction.

Question 9: Patients need transparent notice and practical recourse when AI influences care decisions

RFI question: 9. What challenges within health care do patients and caregivers wish to see addressed by the adoption and use of AI in clinical care? Equally, what concerns do patients and caregivers have related to the adoption and use of AI in clinical care?

Patients and caregivers do not read audit logs. They want to know: “Was the recommendation I received based on the actual guidelines? Can I trust it? And if it was wrong, how do I get it fixed?”

Patient-centered operational needs and concerns

- Clear, understandable notice when AI meaningfully influences a recommendation, documentation, or access decision, including what data types were used at a high level.
- A practical recourse pathway: how to request correction of underlying data, how to contest an AI-influenced recommendation, and how to request human review.
- Ability to access AI-influenced artifacts that are part of the designated record set and to obtain an explanation of how the tool is intended to be used, including known limitations.
- Assurance that data use and retention are minimized and protected, and that access is role-based and logged.

Exhibit 7 summarizes disclosure and recourse elements that can be operationalized for higher-risk AI-enabled workflows.

Exhibit 7. Patient-facing disclosure and recourse elements for higher-risk AI-enabled workflows

Element	What patients/caregivers should receive	Why it matters
Plain-language disclosure of AI involvement	Whether AI was used in generating a recommendation, documentation, or routing decision, and at what point in workflow.	Supports informed trust and avoids deceptive automation.
Purpose and limitations	The intended use and key limitations in plain language, including when not to rely on the tool.	Reduces misuse and supports appropriate escalation.
Recourse and correction path	How to ask questions, request review, and seek correction where appropriate (including documentation correction pathways).	Supports accountability and trust.
Privacy and data use summary	What data are used, whether data are shared externally, and key safeguards.	Supports informed consent-like expectations even where formal consent is not required.
Non-discrimination safeguards (where applicable)	A statement that the organization makes reasonable efforts to identify and mitigate discrimination risk for relevant decision support tools and how concerns can be raised.	Aligns patient-facing transparency with 45 CFR § 92.210 obligations and supports accountability.
Designated record set nuance	When patients inquire about access/amendment, clarify that HIPAA rights depend on whether information is maintained in the designated record set and explain the organization’s process.	Avoids overpromising rights and improves procedural clarity.

Question 10: HHS should prioritize implementation, governance, and reproducible evaluation infrastructure

RFI question: 10. Are there specific areas of AI research that HHS should prioritize to accelerate the adoption of AI as part of clinical care?

HHS research priorities should emphasize implementation and governance, not only model performance. Even strong models can fail in production if integration, monitoring, and update governance are weak.

Exhibit 8 summarizes research areas and support mechanisms referenced in this submission.

Exhibit 8. Research priorities and support mechanisms to accelerate responsible adoption

Priority area	What HHS can do (fund/convene/standardize)	Why it accelerates adoption
Workflow-centric benchmark suites and shared test infrastructure	Fund and convene an open test harness for common clinical workflows that includes provenance-aware writeback and audit export patterns.	Reduces duplicative evaluation burden and enables procurement-ready conformance testing.
Post-deployment monitoring methods ("safety signals")	Fund methods research and implementation science on monitoring dashboards (overrides, data quality, disparity signals, incidents) and operational review cadence.	Makes monitoring feasible and proportional, reducing risk and increasing trust.
Human factors and workflow integration failure modes	Fund pragmatic trials and implementation science on clinician interaction, automation bias, alert fatigue, and documentation integrity.	Addresses the most common deployment failure modes beyond model performance.
Provenance and audit event standardization for AI-influenced actions	Convene standards work on provenance tagging and export patterns for AI-influenced notes/orders and user actions.	Enables retrospective reconstruction and accountable governance across vendors.
Generalizability and external validity evaluation	Fund evaluation research on transportability and site-to-site variability, including subgroup performance and fairness.	Improves confidence in deployed performance across settings.
Nondiscrimination measurement and mitigation in patient care decision support	Support research and guidance prototypes that operationalize "reasonable efforts" to identify/mitigate discrimination, aligned to OCR requirements.	Aligns adoption with civil rights obligations and reduces risk of harmful inequities.
Economic evaluation methods for AI interventions (including transfer effects)	Fund work to standardize AI-specific cost accounting (implementation, maintenance, monitoring, update) and transfer analysis (who pays vs who benefits).	Improves credibility of ROI claims and supports better incentive design.

10a. Published findings on the impact of adopted clinical AI tools

RFI question: 10a. Are there published findings about the impact of adopted AI tools and their use in clinical care?

The published evidence base for deployed clinical AI is growing but remains heterogeneous in study design quality, outcomes measured, and generalizability across sites.

HHS can strengthen adoption by prioritizing reproducible evaluation infrastructure and consistent reporting rather than relying on isolated model-specific claims.

Examples of relevant findings

- A recent scoping review of randomized trials evaluating AI interventions in clinical practice highlights both growth in evaluation activity and persistent variability in study designs and endpoints, reinforcing the need for standardized reporting and workflow-context documentation.
- Ambient documentation studies have reported improvements in clinician time/burden and related measures in specific settings, but outcomes depend strongly on implementation details and governance controls for writeback and review.
- Systematic reviews of online symptom checkers continue to find low and variable diagnostic accuracy, with triage accuracy generally higher but still variable, suggesting that patient-facing tools require careful risk-tiering, disclosures, and monitoring.

HHS can increase the utility of published findings by requiring or incentivizing minimum reporting of tool identity/version/configuration, provenance/writeback behavior, and monitoring signals so that studies are interpretable and comparable across sites.

10b. Literature on costs, benefits, and transfers of using AI in clinical care

RFI question: 10b. How does the literature approach the costs, benefits, and transfers of using AI as part of clinical care?

The economic evaluation literature generally suggests that AI interventions can be cost-effective in specific contexts, but systematic reviews identify recurring methodological gaps and inconsistent reporting of AI-specific cost components and implementation realities.

These gaps directly affect HHS's ability to design incentives that reward real-world value rather than optimistic modeling assumptions.

What the recent systematic review literature emphasizes (examples)

- A 2025 systematic review of cost-effectiveness analyses of AI applications in healthcare reports that many studies conclude AI strategies are cost-effective or cost-saving, but also notes

limitations in modeling assumptions and reporting completeness, including incomplete accounting of implementation and lifecycle costs.

- A 2025 systematic review of economic evaluations for AI-assisted cancer screening/diagnosis highlights “widespread neglect” of AI dynamic learning capability and population differences and reports significant omissions in AI cost components, learning-over-time modeling, and implementation barriers.
- These reviews underscore the importance of AI-specific reporting standards (e.g., CHEERS-AI) and the need to report costs for integration, monitoring, maintenance, retraining/updates, governance operations, and incident response—costs that materially affect net value and transfer effects.

Transfer and incentive considerations (who pays vs who benefits)

In many workflows, costs (integration, monitoring, governance operations) accrue to providers while financial benefits may accrue to payers or may be diffuse, creating adoption barriers under fee-for-service.

HHS should therefore design incentives that (1) support measurable workflow burden reduction and safety monitoring in bounded scenarios, (2) reward auditable governance practices rather than model claims, and (3) reduce inequitable adoption skew toward highly resourced systems.

Reimbursement realism

Where payment policy is needed, CMS should prefer value-based constructs, quality pathways, and model participation terms that require auditable governance artifacts, rather than defaulting to a standalone AI billing code that could create utilization incentives without accountability.

5. Conclusion: Verified, computable clinical reasoning can connect evidence to action at the point of care

HHS can accelerate responsible clinical AI adoption by making accountability feasible in real-world care delivery: risk-tiered governance expectations, interoperable auditability (provenance, user actions, and version/configuration traceability), and incentives for monitoring and evaluation that are proportional to clinical impact and reversibility.

A central distinction in this submission is that deterministic guideline-logic artifacts and probabilistic inference services have different runtime behaviors and should not be treated as interchangeable. That distinction matters for evaluation, monitoring, and reimbursement design. At the same time, both require core safeguards: documentation integrity, nondiscrimination protections, and post-deployment safety monitoring appropriate to risk.

The opportunity is not only to expand AI use, but to improve how evidence is translated into action at the point of care. Clinical guidelines exist. Interoperability standards exist. Coverage pathways often exist. The remaining gap is frequently operational: reliable, reviewable, computable workflow execution that can be audited, monitored, and improved across settings.

Priorities emphasized in this submission

- **Adopt risk-tiered governance expectations** tied to runtime behavior, clinical impact, and reversibility.
- **Strengthen interoperable auditability expectations** for invocation, provenance-aware writeback, user action capture, and runtime version/configuration traceability.
- **Differentiate evaluation and monitoring expectations** for deterministic guideline-logic artifacts versus probabilistic inference services, while preserving shared baseline safeguards.
- **Align incentives to demonstrated workflow and care value** (including monitoring and governance quality), not just tool deployment or unverified performance claims.

Near-term progress is achievable through existing HHS levers, including guidance, ONC certification implementation and testing pathways, OCR implementation guidance for 45 CFR § 92.210, and CMS program terms and quality pathways that reward governable, monitorable deployment.

The infrastructure for safer, scalable clinical AI is already largely in place. The near-term policy task is to align governance, interoperability, and reimbursement so the market rewards systems that can be reviewed, reconstructed, and trusted in practice.

6. Appendices

The appendices below provide working definitions, feasible reporting templates, and source references cited in the body text.

Appendix A. Acronyms and Definitions

A.1 Acronyms

Acronym	Definition
AI	Artificial Intelligence
AHRQ	Agency for Healthcare Research and Quality
ASTP	Assistant Secretary for Technology Policy
CDS	Clinical Decision Support
CDSi	Clinical Decision Support intervention
CDS Hooks	Clinical Decision Support Hooks
CFR	Code of Federal Regulations
CISO	Chief Information Security Officer
CIO	Chief Information Officer
CMIO	Chief Medical Information Officer
CMS	Centers for Medicare & Medicaid Services
CMMI	Center for Medicare & Medicaid Innovation
DSI	Decision Support Intervention
EHR	Electronic Health Record
eCQM	Electronic Clinical Quality Measure
FDA	Food and Drug Administration
FHIR	Fast Healthcare Interoperability Resources
HIPAA	Health Insurance Portability and Accountability Act
HHS	Department of Health and Human Services
HRRP	Hospital Readmissions Reduction Program
IQR	Hospital Inpatient Quality Reporting Program

MIPS	Merit-based Incentive Payment System
NIST	National Institute of Standards and Technology
OCR	Office for Civil Rights
OIG	Office of Inspector General
ONC	Office of the National Coordinator for Health Information Technology
PI	Promoting Interoperability
PHI	Protected Health Information
PII	Personally Identifiable Information
QPP	Quality Payment Program
RFI	Request for Information
SDC	Structured Data Capture
SMART	Substitutable Medical Applications, Reusable Technologies
SSA	Social Security Act
USCDI	United States Core Data for Interoperability
US Core	US Core Implementation Guide

A.2 Definitions

The working definitions below are included as provided in the source submission.

Term	Definition
Clinical AI / decision support	Software-enabled methods that generate recommendations, predictions, summaries, or workflow actions used in clinical care, including both deterministic and probabilistic approaches.
Deterministic guideline-logic artifact	A deterministic, version-controlled artifact that applies defined guideline/policy logic to inputs to produce repeatable outputs for the same inputs and configuration.
Probabilistic inference service	A service that produces outputs based on statistical inference, often sensitive to data distributions and subject to performance variation over time and across populations.
Formal verification (bounded)	Verification of internal logical properties of a deterministic artifact over defined states (e.g., consistency, reachability, completeness), which does not establish clinical benefit or appropriateness across contexts.
AI audit artifact package	A minimum set of artifacts (identity, version/config, provenance/writeback behavior, user action capture, limitations, monitoring plan, change logs) sufficient to support retrospective reconstruction, governance review, and incident response.
Provenance	Machine-readable representation of who/what generated information and how it was derived, used to identify AI-influenced content and enable traceability.
Post-deployment safety monitoring	Ongoing monitoring for safety signals proportional to risk, including data quality/missingness, overrides/escalations, disparate application, incidents, and update impacts (and drift monitoring for probabilistic tools where appropriate).
Risk tier	A qualitative classification driven by clinical impact, autonomy, patient-facing use, record writeback, scale, reversibility, and discrimination risk triggers.

Appendix B. Lightweight Evaluation and Reporting Template (feasible minimum)

The fields below are intended to be feasible for developers and deployers to complete and to support procurement review, monitoring, and retrospective analysis without disclosing proprietary model internals.

Field	Minimum content to report
Tool identity	Name, developer, intended use, clinical setting, user roles
Risk tier justification	Trigger-based rationale (clinical impact, autonomy, patient-facing, writeback, reversibility, discrimination risk)
Runtime behavior type	Deterministic guideline-logic artifact vs probabilistic inference service (or hybrid)
Version/configuration	Version, configuration parameters, enabled features, update policy
Data dependencies	Data types used, required fields, missingness sensitivities, data quality checks
Output and use	Output type, how clinicians should interpret/use outputs, escalation/override processes
Evaluation endpoints	Workflow endpoints, safety endpoints, subgroup/fairness endpoints where relevant
Monitoring signals	Override rates, data quality/missingness, disparity signals, incident tracking, update impacts; drift/performance monitoring where appropriate
Governance operations	Review cadence, responsible roles, incident response process, regression testing (including guideline/version updates)
Patient-facing elements (if applicable)	Disclosure, recourse, privacy summary, nondiscrimination safeguards communication

Appendix C. AI Audit Artifact Package (suggested minimum field set)

This package is intended to be stored and accessible for governance and retrospective review. For higher-risk use cases, it should be reviewable by the deploying organization's clinical governance and compliance functions and support patient recourse requests.

Artifact category	Expanded minimum content (portable across vendors/sites)
Identity and intended use	Intended use statement; contraindications; workflow description; user role assumptions
Versioning and configuration	Artifact/model version; configuration; prompt templates where applicable; feature flags; update cadence
Data lineage and dependencies	Input data types; mappings; missingness sensitivities; pre-processing; source systems
Provenance and record writeback	Writeback targets; provenance attachment method; separation of AI-suggested vs clinician-authored content
User interaction capture	Accept/override/modify; escalation triggers; audit export capability
Limitations and failure modes	Known limitations; subgroup considerations; robustness notes; security/privacy considerations
Verification and testing evidence	Deterministic: internal logical property checks + regression tests; Probabilistic: validation approach + robustness checks
Monitoring plan	Signals, thresholds, review cadence, roles, incident response, documentation updates
Nondiscrimination documentation (if applicable)	Reasonable-efforts documentation aligned to 45 CFR § 92.210; monitoring and mitigation approach
Transfer/payment considerations (where relevant)	Workflow cost accounting approach; who bears costs vs who benefits; documentation for program reporting if required

Appendix D. References

Note: Full URLs are provided in this section only. Citations in the body refer to these numbered references.

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[11] eCFR. Information Blocking Regulations (45 CFR Part 171).
<https://www.ecfr.gov/current/title-45/subtitle-A/subchapter-D/part-171>

[12] eCFR. HIPAA Right of Access (45 CFR 164.524).
<https://www.ecfr.gov/current/title-45/subtitle-A/subchapter-C/part-164/subpart-E/section-164.524>

[13] eCFR. HIPAA Right to Request Amendment (45 CFR 164.526).
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Appendix E. Candidate CFR Citations for Legal Review

The CFR citations below are provided as candidate anchors to support counsel validation and to increase responsiveness to the RFI's request for "specific changes" with "applicable CFR citations."

This appendix is not legal advice and should be reviewed and finalized by counsel for any formal compliance interpretation.

ONC / ASTP / interoperability and certification context (candidate anchors)

- 45 CFR 170.315(b)(11) (Decision Support Interventions) and related ONC Certification Program provisions supporting transparency and intervention risk management concepts.
- Associated ONC test methods and implementation guidance as the primary operational mechanism for making auditability and export expectations testable.

OCR / nondiscrimination (candidate anchors)

- 45 CFR § 92.210 (patient care decision support tools), including requirements prohibiting discrimination through use of such tools and the duty to make reasonable efforts to identify and mitigate discrimination risk where tools employ input variables measuring protected characteristics.

HIPAA Security Rule (candidate anchors)

- 45 CFR 164.312 and related Security Rule provisions addressing audit controls and security safeguards, implemented through documented policies rather than fixed retention-duration mandates.

Information blocking (candidate anchors)

- 45 CFR Part 171 (information blocking regulations), implemented with appropriate reliance on the exceptions framework (e.g., privacy, security, infeasibility, fees, licensing), rather than implying blanket access rights.

CMS (candidate anchors for counsel validation; mechanism-level proposals in Section 4.2)

- MIPS / Quality Payment Program: 42 CFR Part 414, Subpart O (Quality Payment Program framework).
- Promoting Interoperability (PI): 42 CFR § 414.1375 (Promoting Interoperability performance category).
- Improvement Activities (IA): Candidate anchor within Part 414 Subpart O for Improvement Activities performance category (e.g., provisions addressing IA scoring/requirements), to be validated by counsel for the specific mechanism used.
- Hospital quality reporting (IQR): 42 CFR § 412.140 (Hospital Inpatient Quality Reporting (IQR) Program).